CCCA Cell-State Analysis — Wang et al. 2019 (Glioblastoma)

IDH-wildtype glioblastoma · 10x · 9,634 malignant cells (7 samples) · non-branching developmental axis

GO / NO-GO GATES

GATE 1 · Integration & continuity — GO

Judged on the embedding + lobe composition (Kürten fork: never a scalar). Harmony on sample cut the same-sample neighbour fraction 0.99→0.6 and dissolved the theta=2 monochrome patient lobes: one connected component, no lobe a single patient (dominant share ≤0.35; 6–7/7 samples per lobe). Mixing is partial (iLISI median 1.9/7) — expected GBM CNV identity within a continuous multi-sample manifold, not an archipelago. GO.

GATE 2 · Module pattern (non-branching) — GO

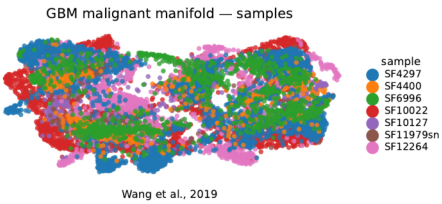
PPT tree over-branches a continuous cloud (forks 5→18 with node count, none stable), and the two arm-candidates Gast & pSom co-vary (spatial r=+0.16) — no bifurcation. Shared Gast/pSom gradient with neural crest as the inverse terminal end (Crest–pSom r=−0.87; Crest-high decile at pseudotime 0.84). Non-branching pattern present. GO.

GATE 3 · Hub cross-repression — NA

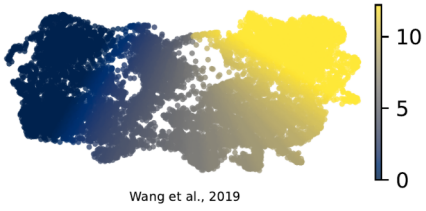
Not applicable — the manifold is non-branching (single principal curve, 0 forks), so there is no hub bifurcation on which to test mutual Gast↔pSom cross-repression.

STEP FIGURES

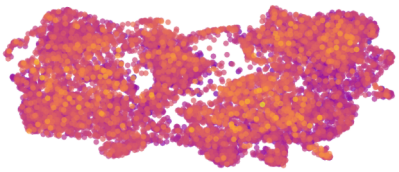
Samples (Gate 1: batch integration)



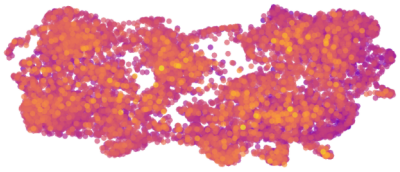
Pseudotime trajectory (principal curve)



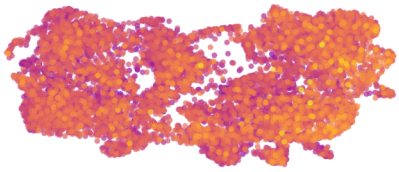
Gastrulation signature



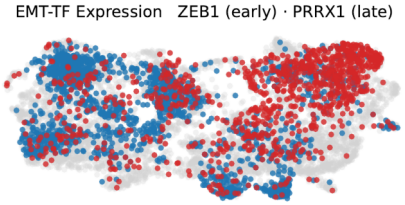
Somitogenesis signature



Neural-crest signature



EMT-TFs ZEB1 (early) / PRRX1 (late)



KEY PARAMETERS & RATIONALE

Threshold	min 237 cells/sample (largest multiplicative gap 396→143, 2.8×) → 7/10 samples, 9,650 cells
Integration	Harmony theta=10 on sample; n_pcs=20 (scree + 10x precedents); UMAP spread=1.3; overdispersed=2,113
Islands	16 detached cells removed (radius graph) → 9,634 on one connected component
Geometry	Linear-pSom/Crest — non-branching principal curve (ElPiGraph Nodes=20, epg_lambda=0.3, epg_mu=100)
	fold-free (self-approach 2.8, stable across Nodes 15/20/30); PPT tree 0 stable forks
Root	curve tip 2 — Gast/pSom-high, Crest-low developmental origin (neural crest is late → terminus)
Scope	whole manifold kept — cell cycle orthogonal to developmental axis (|corr|<0.1; uniform cycling/lobe)
EMT-TFs	derived ZEB1 (root, 70% expr) → PRRX1 (terminus, 40% expr); Snail/Twist/FOXC <5% detection
VERDICT	**Pattern reproduces (non-branching): one continuous gastrulation/somitogenesis → neural-crest developmental axis, EMT-TFs ZEB1→PRRX1. Gate 1 GO · Gate 2 GO · Gate 3 NA.**